

ADVERSARIAL SMART CONTRACT REVIEW

Flap

template audit.

Subject	FlapTaxTokenV3 + Portal template (flap.sh)
Portal	0xe2cE...9De0
Token impl (TaxedV3)	0x024f...6422
Network	BNB Chain · chainId 56
Compiler	Solidity 0.8.24
Adversaries	8
Rules applied	2,184



0

CRITICAL

2

HIGH

3

MEDIUM

4

LOW / INFO

EXECUTIVE SUMMARY

A tax-token launchpad with a vault attachment surface.

This is a **template audit**. Every token deployed via Flap's Portal shares the same FlapTaxTokenV3 implementation, the same bonding-curve math (selectable via CurveType), and the same DEX-migration path. Pentagonal has reviewed the template once; subsequent tokens inherit that review.

Flap is structurally similar to Four.meme — a bonding-curve memecoin launchpad on BSC that graduates to PancakeSwap — with two notable additions: **asymmetric buy/sell tax rates** baked into the token, and **vault-attached tokens** where an external vault receives the tax stream and may itself be anything from a staking pool to a treasury to a wallet the deployer controls. Tokens use a **vanity suffix 7777**.

The template **passes** four of the eight adversaries (no reentrancy vector in the Portal's trade path, no oracle dependence, no overflow risk in the curve math, arithmetic checked throughout). It **flags** for the remaining four — detailed below.

Findings overview

SEVERITY	COUNT	NOTES
CRITICAL	0	No drain vector exists in the template itself.
HIGH	2	Asymmetric tax mutability, vault trust surface.
MEDIUM	3	Migration race, dispatch DoS, curve parameter range.
LOW / INFO	4	Standard tax-token surface with documented additions.
Total	9	—

Adversary scorecard

ADVERSARY	STATUS	FINDINGS
Reentrancy Hunter	✓ Clear	0
Flash Loan Attacker	✓ Clear	0
Access Control Prober	⚠ Findings	1 (high)
Overflow Saboteur	✓ Clear	0
Oracle Manipulator	✓ Clear	0
MEV Predator	⚠ Findings	1 (medium)
Economic Exploit	⚠ Findings	4
Gas Griefer	⚠ Findings	1 (medium)

The two things to verify on every Flap token.

F-01 · Asymmetric buy/sell tax rates with admin mutability

HIGH

Found by **Economic Exploit** + **Access Control Prober**

FlapTaxTokenV3's headline feature is asymmetric tax — the token can ship with, e.g., 1% buy tax and 99% sell tax. This is not a bug; it's a parameter. But the same admin path that sets the initial rates can *update* them while the token sits on the bonding curve, which means a token marketed as "1%/1%" can become "1%/99%" between your buy and your sell with zero on-chain warning.

Some tokens' rates renounce automatically post-graduation. Many do not. Treat any pre-graduation Flap token's sell tax as a *current reading*, not a guaranteed cap.

What it means for you. Always read `getTokenV8Safe(token).sellTaxRate` right before any meaningful exit. Off-chain price aggregators (GoPlus, DexScreener) often miss tax-token mechanics and report 0% even when on-chain returns 300+ bps.

F-02 · Vault attachment is a trust surface, not a security feature

HIGH

Found by **Economic Exploit**

Flap's Vault Portal lets the deployer bind an arbitrary contract (vault) as the destination of tax revenue. Any contract — a real staking pool, a multisig, or a wallet the deployer controls. The *existence* of a vault doesn't say anything about who controls the funds in it.

What it means for you. Before assigning credibility to a vault-attached token, fetch the vault address (Flap REST API or Portal) and audit the vault contract itself. A vault that's a 1-of-1 EOA is the deployer with extra steps.

Migration, dispatch, and the curve parameter space.

F-03 • Graduation migration is a known MEV target

MEDIUM

Found by **MEV Predator**

Flap's V2 migrator (used for tax tokens) adds liquidity to PancakeSwap in a transaction that's predictable from the Portal's events. The first block after migration is consistently sandwich-targeted; retail buyers in that window typically eat 1–5% slippage to MEV bots.

F-04 • Tax dispatch can be grieved if the vault reverts

MEDIUM

Found by **Gas Griefer**

`TaxProcessor.dispatch()` forwards the accumulated tax to the bound vault. If the vault is malicious and reverts on `receive()` / `tokenReceived` / similar hooks, dispatch fails and tax accumulates indefinitely in the processor. The token continues trading; the vault just doesn't get paid. Holders aren't drained, but accounting drifts.

F-05 • Curve parameter surface is wide

MEDIUM

Found by **Economic Exploit**

Flap supports many `CurveType` variants (`CURVE_RH_BNB`, `CURVE_RH_USD`, etc.) plus arbitrary `r/h/k` parameters. Most behave correctly, but extreme curves can be configured to make early-buyer dumps look like rugs even without admin action. A flat curve in particular concentrates impact at the upper end of the bonding range.

Background notes.

- **F-06** · Standard tax-token surface; no surprises beyond F-01.
- **F-07** · No reentrancy vectors in the Portal's trade path.
- **F-08** · Arithmetic is checked throughout; no unchecked blocks of concern in the V3 implementation.
- **F-09** · Black-hole address (`0x00576E...0DEad`) is used for burns and is honored correctly across the trade path.

RECOMMENDATION

How to size and where to use this template.

The Flap template is **technically sound** at the contract level. There is no critical drain vector, no reentrancy exposure, and no oracle-manipulation surface. Risks are **economic and operational**:

1. Buy/sell tax rates can be modified pre-graduation and can be asymmetric by design.
2. The vault attachment is a trust surface, not a security feature.
3. Migration is sandwich-targeted.
4. Curve parameters span a wide range; verify the curve type before sizing.

For **short-duration speculative exposure** with continuously-checked tax rates and a small position size, the template is acceptable. For **long-term holding** — treasury, savings, payment rails — Flap tokens are inappropriate without a vault audit and confirmed rate renunciation post-graduation.